

Independent co-located validation cells for continental-scale arsenic screening: cross-survey agreement and a machine-actionable disagreement queue from harmonized open geochemical data

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Abstract

Machine-learning global arsenic hazard maps explicitly identify sparse independent ground truth as their limiting factor. We show that harmonized open continental surveys can supply a quality-controlled, provenance-chained validation point set at near-zero marginal cost. Within a 191,715-record harmonized geochemical database we identify 0.1°-grid cells in which two *independent* European surveys — GEMAS (Geochemical Mapping of Agricultural and Grazing Land Soil; sampled 2008–2009, inductively coupled plasma mass spectrometry, ICP-MS, after aqua-regia digestion) and FOREGS (Forum of European Geological Surveys; sampled 1997–2001, separate sample network and laboratories) — both report soil arsenic: $n=26$ co-located cells. Despite different decades, media conventions, digestion protocols and laboratories, the surveys agree strongly: Spearman $\rho = 0.681$ ($p = 1.3 \times 10^{-4}$), median GEMAS/FOREGS ratio 0.982 — replication to within 2% at the median — and 19 of 26 cells (73%) agree within a factor of two. Both surveys jointly confirm elevated cells (two cells above 20 mg/kg in both, one above 45 mg/kg in both), while the harmonized screening layer flags 688 of 8,979 soil-As records (7.7%) above 20 mg/kg and 162 (1.8%) above 45 mg/kg across four continents. The seven cells outside factor-two agreement form an automatically generated *disagreement queue* that prioritizes re-sampling, headed by a Brittany cell where GEMAS reports 208.5 versus FOREGS 59.6 mg/kg. The deliverables — a confidence-scored validation point set, the disagreement queue, and a global joint-coverage map showing where no independent validation exists — are published as machine-readable layers for risk-map builders. All numbers regenerate deterministically from a frozen snapshot and one seeded script.

1 Introduction

Global arsenic hazard mapping has moved to machine learning: Podgorski and Berg trained a groundwater-As hazard model on tens of thousands of measurements and projected it worldwide, while stating plainly that sparse, uneven ground truth — especially *independent* data not used in training — limits both validation and refinement [3]. The soil pathway shares the problem: continental surveys exist on several continents [5, 4, 6], but each was built by one consortium with one protocol, so a model validated against a survey is partly validated against that survey’s idiosyncrasies.

There is, however, an under-used property hiding in the open data: where two independent surveys sampled the *same place*, their agreement (or disagreement) is itself a measurement —

of survey reliability, of temporal stability, and of the trustworthiness of any single point used as ground truth. Extracting this signal requires the surveys to live in one schema with common units, coordinates and quality flags; that is precisely what a harmonized multi-survey database provides mechanically.

This paper extracts the arsenic co-location signal from two European surveys and packages it for consumption by risk-map builders. Contributions: (1) the co-located cell set (GEMAS $\times$ FOREGS at 0.1° ; $n=26$) with per-cell agreement statistics — to our knowledge the first systematically derived cross-survey validation point set for European soil As; (2) quantified cross-survey reliability ($\rho=0.681$, median ratio 0.982, 73% within factor two) — a number neither survey can produce alone; (3) an automatically generated disagreement queue for re-sampling, with full provenance to source rows; (4) a global joint-coverage panel across 8 sources and four continents showing where independent validation is impossible today, turning coverage gaps into acquisition targets. The topic was proposed automatically by the anomaly-replication tier of an autonomous geochemical data skill after its screening layer flagged recurring As exceedances; this paper is the validation-side complement.

2 Materials and methods

2.1 Sources and harmonization

All records derive from one frozen snapshot (`retest-acquisition-coverage-v12-6c221cf`) of a harmonized open geochemical database (191,715 determinations, 29 sources; concentrations normalized to mg/kg, coordinates to the World Geodetic System 1984, WGS84; method families and per-record provenance hashes retained). The soil-As slice comprises 8,979 records from 8 sources on four continents (Table 2). The two co-location surveys are GEMAS [4] (agricultural/grazing soils, ICP-MS after aqua regia) and FOREGS [5, 1] (natural topsoil and subsoil, independent network and laboratories). The surveys differ in sampling decade, media convention and digestion protocol; agreement statistics therefore measure *operational* reproducibility — the quantity a risk-map builder actually experiences when treating a survey value as ground truth.

2.2 Co-location and statistics

Records are aggregated to 0.1° cells (roughly 11 km at these latitudes, comparable to the resolution of global hazard maps [3]) by the median of each survey’s records in the cell; cells where both surveys report As form the validation set ($n=26$). Agreement is summarized by the Spearman rank correlation ρ , the median GEMAS/FOREGS ratio, and the count within factor-two agreement ($|\log_2 \text{ratio}| \leq 1$). Screening thresholds of 20 and 45 mg/kg are used to count exceedances; these span common European soil guidance ranges and serve here purely as screening values, not regulatory conclusions. All statistics are deterministic (seed 20260819 where resampling is involved) and regenerate from the snapshot.

3 Results

3.1 Independent surveys agree to 2% at the median

Figure 1 and Table 1 present the full validation set. Across 26 cells spanning 1.8–208.5 mg/kg, Spearman $\rho = 0.681$ ($p = 1.3 \times 10^{-4}$), the median GEMAS/FOREGS ratio is 0.982, and 19/26 cells (73%) agree within a factor of two. For context, the same-sample method offset between aqua-regia

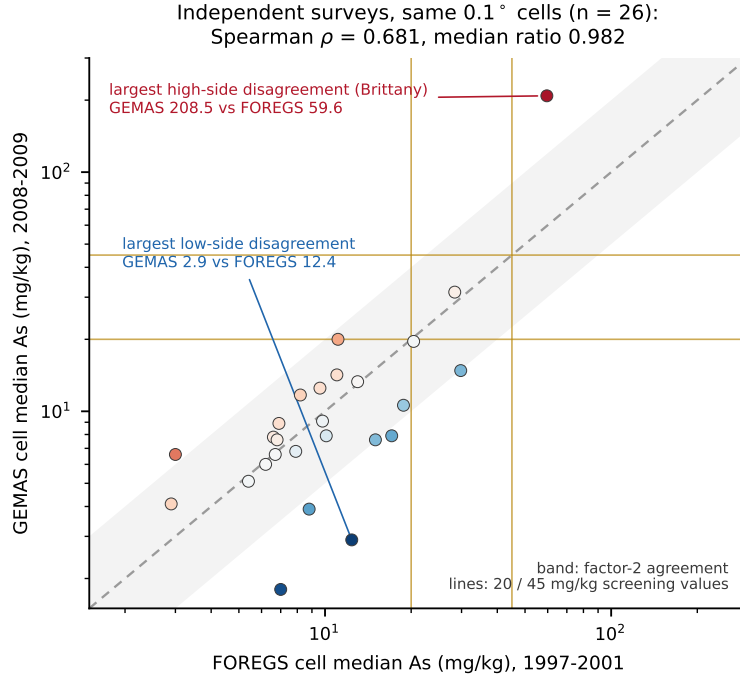


Figure 1: GEMAS versus FOREGS cell-median soil As in the 26 co-located 0.1° cells (log-log). Grey band: factor-two agreement; gold lines: 20 and 45 mg/kg screening values; point color: $\log_2$ (GEMAS/FOREGS). The largest high-side and low-side disagreements are annotated and head the re-sampling queue (Table 1, top and bottom).

and total As determinations within one survey is 1.23 at the median [2]; the cross-survey cell-median ratio of 0.982 indicates that spatial aggregation plus independent sampling does not add systematic bias beyond the method layer — the agreement one would hope for, now quantified.

3.2 Joint exceedance confirmation and the disagreement queue

Both surveys independently place two cells above 20 mg/kg: the Brittany cell (also the only cell above 45 mg/kg in both surveys) and an Alentejo, Portugal cell (39.03°N 7.67°W : GEMAS 31.5, FOREGS 28.4 mg/kg, agreeing within 11%) — high-As cells confirmed by two consortia a decade apart carry validation weight no single survey can provide. The seven cells outside factor-two agreement (Table 1, *) are emitted as a re-sampling queue with full provenance. The queue head, a Brittany cell (47.62°N 2.31°W), shows GEMAS 208.5 versus FOREGS 59.6 mg/kg — both clearly elevated, disagreeing 3.5-fold in magnitude; candidate explanations (local heterogeneity at sub-cell scale, media difference, a genuine change between 1998 and 2008) are distinguishable only by new sampling, which is the point of the queue. The low-side tail (five cells where FOREGS exceeds GEMAS 2–4-fold, concentrated in Iberia and southern Italy) suggests media or digestion effects worth targeted follow-up rather than a random scatter.

3.3 The global validation desert

Figure 2 and Table 2 put the 26 cells in global context: 8,979 harmonized soil-As records exist across Europe, North America, Africa and South America, yet *every* co-located independent pair lies in

Table 1: The complete validation point set: all 26 co-located 0.1° cells, sorted by GEMAS/FOREGS ratio (descending). Cell medians in mg/kg. Cells outside factor-two agreement (ratio > 2 or < 0.5) constitute the disagreement queue (7 cells, marked *).

Lat	Lon	GEMAS	FOREGS	Ratio	Lat	Lon	GEMAS	FOREGS	Ratio
47.62	-2.31	208.5	59.6	3.50*	37.13	-5.41	6.0	6.2	0.98
37.32	14.66	6.6	3.0	2.23*	39.42	-5.32	19.6	20.4	0.96
44.66	6.59	20.0	11.1	1.81	37.21	21.98	5.1	5.4	0.95
48.42	12.83	11.7	8.2	1.43	51.98	1.04	9.1	9.8	0.93
38.67	-0.55	4.1	2.9	1.41	50.92	11.42	6.8	7.9	0.86
48.72	6.99	12.5	9.6	1.29	50.40	4.07	7.9	10.1	0.78
42.39	11.84	14.2	11.0	1.29	50.42	12.17	10.6	18.8	0.56
49.67	7.92	8.9	6.9	1.29	43.75	-0.20	7.6	15.0	0.51
50.34	2.68	7.8	6.6	1.18	47.24	11.89	14.8	29.8	0.50*
46.71	6.67	7.6	6.8	1.12	37.09	22.43	7.9	17.1	0.46*
39.03	-7.67	31.5	28.4	1.11	40.71	15.45	3.9	8.8	0.44*
47.78	5.71	13.3	13.0	1.02	41.12	0.83	1.8	7.0	0.26*
35.20	26.08	6.6	6.7	0.99	40.90	-7.55	2.9	12.4	0.23*

Table 2: Harmonized soil-As screening panel by source (records passing coordinate quality control, QC). Median in mg/kg; counts above the 20 and 45 mg/kg screening values. A further 2,842 soil-As records (AfSIS Africa 1,941; Yangtze basin 821; Ningbo 80) exist in the snapshot but carry no per-record coordinates and are excluded here by coordinate QC — itself a documented gap.

Source	n	Median	> 20	> 45
GEMAS Europe [4]	4,448	6.0	315	68
USGS CONUS soil [6]	1,977	4.9	39	6
FOREGS topsoil [5]	892	9.5	159	40
FOREGS subsoil [5]	838	9.0	147	39
PANGAEA Barents C-horizon	594	0.6	2	0
PANGAEA Amazonas	166	6.6	16	4
PANGAEA NE Brazil	21	10.4	4	2
PANGAEA North Africa	43	7.5	6	3
Total	8,979		688	162

Europe — the only region where two independent continental surveys overlap. The USGS (U.S. Geological Survey) CONUS survey [6] contributes 1,977 records (median 4.9 mg/kg, 39 above 20 mg/kg) but has no independent partner survey in the harmonized corpus; Africa and South America are covered by single sparse campaigns. In the terms of [3], the map of where validation *could* happen is itself the acquisition agenda: any new open survey overlapping an existing one immediately mints validation cells at zero laboratory cost.

4 Discussion

What the point set is for. A global hazard model can use the 26 cells three ways: as held-out validation points whose *pairing* quantifies label noise (the factor-two band is an empirical error bar on any single survey value); as calibration anchors where both surveys agree; and as abstention flags where they disagree. Because every cell links to its source rows by provenance hash, a model

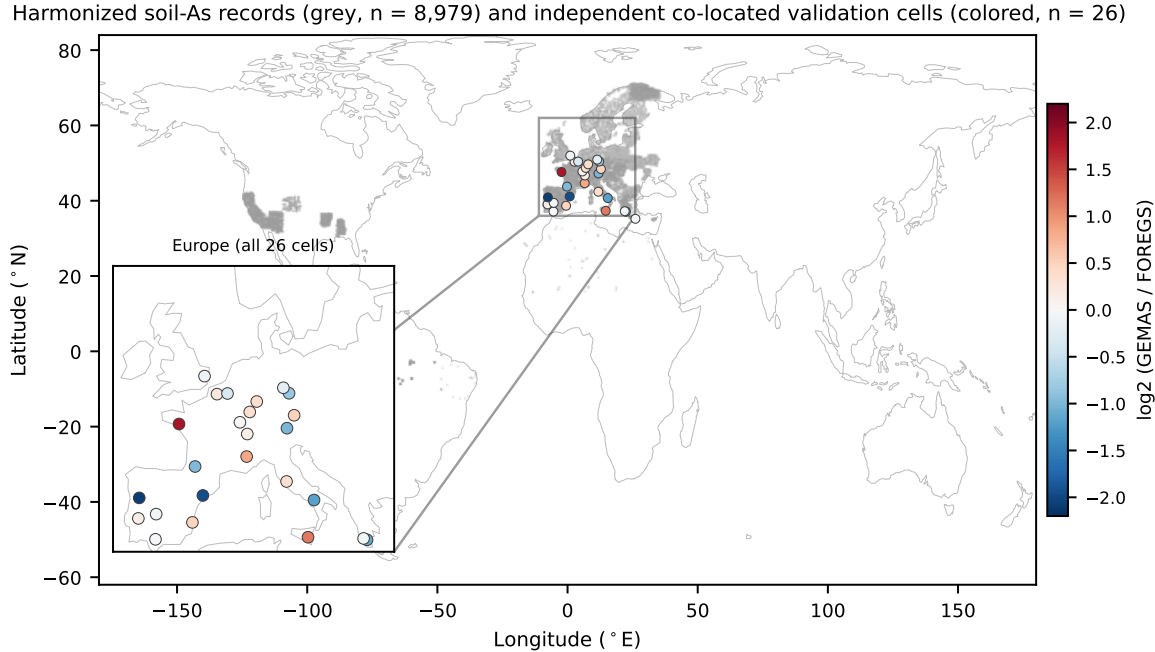


Figure 2: Global joint-coverage panel. Grey: all 8,979 harmonized soil-As records (8 sources, four continents). Colored circles: the 26 co-located validation cells — all in Europe (inset), colored by $\log_2$ (GEMAS/FOREGS). Land outline: Natural Earth.

builder can trace any surprise to the original laboratory report — the property [3] identify as missing from most compiled training sets. A fourth use is negative: the coordinate-QC exclusion of the African AfSIS panel (§3.3) shows that some published data can never mint validation cells as released, which is itself actionable feedback to data producers.

Agreement is a database product, not a survey product. Neither consortium designed for cross-validation; the signal exists only after units, coordinates, media labels and method families are harmonized. The 73%-within-factor-two number is thus an argument for harmonization infrastructure in general: it was recoverable from data that had been public for over a decade.

Limitations. Twenty-six cells is a pilot-scale validation set; 0.1° aggregation trades spatial precision for pairing count, and sub-cell heterogeneity contributes to the observed spread. GEMAS and FOREGS differ in media and digestion, so the ratio statistics fold method effects into operational reliability (deliberately — but a same-protocol replication would be cleaner). The soil pathway complements but does not substitute for groundwater measurements, where the exposure burden of [3] is concentrated. Finally, 2,842 soil-As records (including the entire AfSIS African panel) fail coordinate QC and cannot enter any spatial product; they are retained in the database with explicit missing-coordinate flags rather than dropped, and recovering their georeferencing from source documentation is queued as an acquisition task.

New questions generated. (Q1) Do the five low-side disagreement cells share a media or digestion signature that a targeted 10-sample campaign could isolate? (Q2) Which planned or ongoing surveys, if released openly, would create the first non-European validation cells — and can their sampling plans be nudged (at the margin) toward existing survey sites, where each site doubles as a validation cell? (Q3) At what aggregation scale does cross-survey agreement peak — i.e., what is the natural spatial support of “ground truth” for continental soil As?

5 Conclusions

Two independent European surveys, harmonized into one schema, agree on co-located soil arsenic to 2% at the median and within a factor of two in 73% of cells — and their seven disagreements form a concrete, provenance-chained re-sampling queue headed by a 3.5-fold discrepancy in Brittany. The validation point set, disagreement queue and joint-coverage map are published as machine-readable layers; the global map makes the validation desert outside Europe explicit and actionable. Independent ground truth for arsenic screening does not need to be commissioned — some of it is already latent in open data, waiting to be harmonized out.

Data and code availability

Input surveys: GEMAS [4], FOREGS [5, 1], USGS CONUS [6], plus AfSIS and PANGAEA campaign datasets as listed in Table 2, ingested with per-record licenses and provenance hashes. The frozen snapshot identifier, analysis script (`p345_full.py`, seed 20260819), result JSON (`p345_full_results.json`) and figure sources are archived with SHA-256 receipts in the pipeline’s `research-products/paper-demo-20260817` directory.

Agent disclosure

This draft, including analysis design, statistics, figures and text, was produced end-to-end by an AI research agent operating the Global Geochemical Atlas skill; the topic was proposed by the skill’s screening and discovery layers and has not yet completed human review. Screening-level framing is intentional and load-bearing.

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